Spectral Shrunk Clustering: Reproduction, Numerical Audits, and Subset Uncertainty

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Abstract

Spectral Shrunk Clustering (SSC) refines a spectral representation through a convex fidelity–fusion objective. We examine its mathematical target, reproduce available benchmark settings, and distinguish optimization validity from final clustering accuracy. Missing historical sample indices motivate a predeclared source/subset sensitivity campaign. Joint ACC/NMI matching replaces ACC-only retrospective selection; mean/sample-SD and native-objective-selected assignments remain separate endpoints. Numerical audits preserve feasible dual bounds and correct a graph fragmentation statistic. Source and implementation repairs include native 16-by-16 MSRA data and explicitly labelled LDA adaptations. These experiments quantify reproducible reconstructions and limitations, without identifying historical datasets merely from numerical agreement.

1 What SSC optimizes

Spectral clustering (SC) represents samples using graph-Laplacian eigenvectors and then applies K-means. Chang et al. [Chang et al., 2015] argue that directly clustering the initial spectral coordinates can be unreliable, and propose learning *shrunk patterns* that preserve their anchors while contracting neighboring representations. Their contribution is a convex representation-refinement formulation, relaxation of an orthogonality constraint in this stage, and an iterative algorithm with clustering experiments. SSC here means Spectral Shrunk Clustering, rather than Sparse Subspace Clustering.

The input-space affinity A determines $L_A = I - D_A^{-1/2} A D_A^{-1/2}$. The c smallest eigenvectors form $F \in \mathbb{R}^{n\times c}$. A second affinity W is constructed from F and controls refinement (Figure 1). With each undirected edge stored once in incidence matrix B, Equation 8 is

$$P(G) = \sum_i \|g_i - f_i\|_2 + 2\gamma \sum_e w_e \|(BG)_e\|_2. \tag{1}$$

The factor 2 implements the paper's ordered-pair sum. Its stationary IRLS system uses $S + 2\gamma\widetilde{L}$ (E8), whereas printed Equation 13 uses $S + \gamma\widetilde{L}$ (E13). Thus E13 at nominal gamma targets E8 at effective gamma/2. We retain both tracks, rather than calling unequal targets a solver discrepancy.

For fixed F, W, nonnegative weights make (1) convex; unsquared fidelity does not guarantee uniqueness. Graph construction, eigenspace selection and final K-means are separate operations. A globally optimal G therefore need not improve ACC or NMI. Orthogonal eigenspace rotations preserve Euclidean distances, but a changed cutoff eigenspace or row normalization can alter clustering.

Preprint.

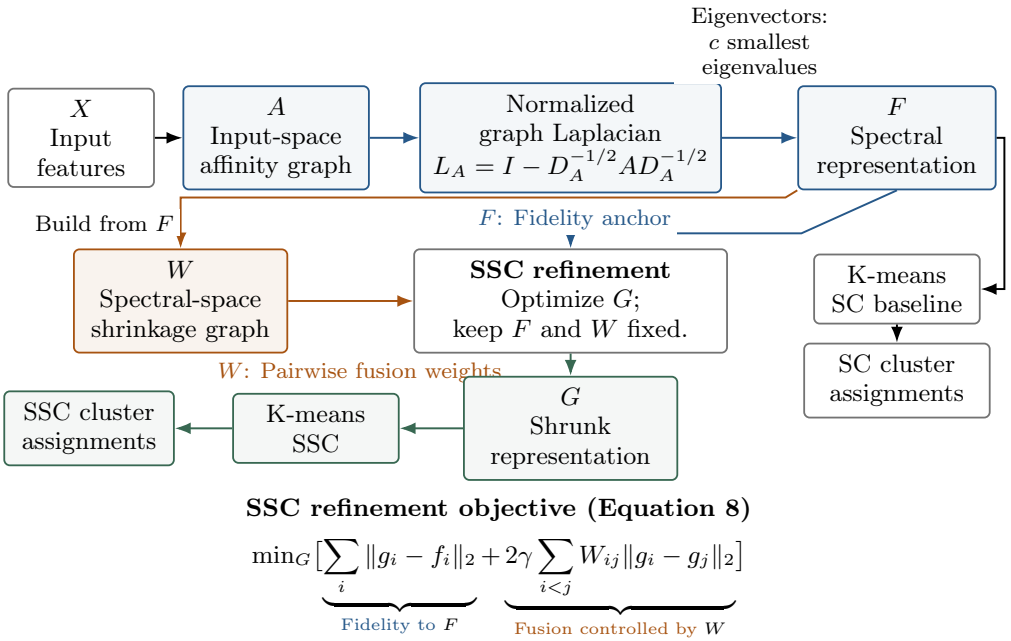


Figure 1: The two graph roles are distinct. A determines the initial spectral geometry F; W, built in that geometry, controls fusion while F remains fixed. Matched SC clusters the same F, and SSC clusters G.

2 Published results and the reproduction protocol

We verified all 288 reported means and SD values in the paper's Tables 1–2: nine methods, eight benchmarks and two metrics. The paper states $k = 5$, 50 random initializations, a five-value parameter grid and native-objective selection. It does not fully resolve the SD hierarchy, source indices, Gaussian width or graph symmetrization. Our choices are union kNN, median positive directed-neighbor squared-distance width, deterministic eigensolver initialization, and explicit one-start Lloyd rounding. ACC uses optimal label assignment and NMI uses geometric entropy normalization.

The paper's SSC ACC gains do not imply superiority on every metric: YaleB SC/SSC have ACC 34.8/49.8 but NMI 55.6/46.8. We therefore minimize joint discrepancy in percentage points,

$$D_{\max} = \max(|\overline{\mathrm{ACC}} - \mathrm{ACC}_p|, |\overline{\mathrm{NMI}} - \mathrm{NMI}_p|), \qquad D_{\mathrm{mean}} = \tfrac{1}{2}(d_{\mathrm{ACC}} + d_{\mathrm{NMI}}).$$

We rank by $D_{\max}$, then D_{mean}, with a 10^{-12} tolerance and frozen compatibility tie breaks. Published-grid gamma values are not penalized for differing from 1. Both metrics and SD come from one configuration and its complete 50 seeds. The single native-best-of-50 assignment has no across-run SD; existing independent batches are a separate repeatability endpoint. Raw LDA inertia across learned subspaces is not treated as a common selection objective.

2.1 Dataset-subset uncertainty and retrospective numerical matching

Original sample indices were unavailable. We evaluated predeclared candidate subsets and preparation protocols and selected the smallest joint discrepancy from published ACC/NMI. **This is a retrospective numerical match, not label-free selection or an unbiased superiority test.** The selected manifests are reproducible; their identity as the authors' historical datasets remains unverified. AR seven-of-fourteen rules, UMIST deduplication/omission hypotheses, and BinAlpha 30-per-letter draws preserve explicit source indices. Different membership always rebuilds A, F, W, G. Diagnostic parents and native 16-by16 MSRA are separately labelled; resolution is not recovered by upsampling. ORL/JAFFE membership is not arbitrarily searched. Yale source, resizing and Y0–Y3 normalization factors remain distinct. USPS uses the full 9,298 rows in specification-compatible experiments.

Data	Paper ACC	Paper NMI	Our ACC	Our NMI	$D_{\max}$	Fit
ORL	76.00 ± 1.60	88.60 ± 1.50	74.63 ± 0.64	89.71 ± 0.23	1.37	I
JAFFE	81.20 ± 1.60	84.30 ± 1.60	74.75 ± 8.37	85.60 ± 4.32	6.45	T
YaleB	49.80 ± 1.60	46.80 ± 1.30	42.64 ± 0.61	52.19 ± 0.31	7.16	T
USPS	75.50 ± 1.90	79.80 ± 1.60	73.87 ± 7.53	82.84 ± 3.96	3.04	I

Table 1: Historical SSC scalar records, reselected jointly by $D_{\max}$ then D_{mean}; this is a retrospective match, not label-free selection. Means $\pm$ sample SD use one configuration and all 50 seeds. I: iterate stopping without a gap certificate; T: iteration cap. None of these historical G arrays is recoverable, so current solver audits cannot certify their exact outputs. Settings and source versions appear in the appendix.

The frozen catalogue contains 4736 intended entries. New execution coverage has not yet been aggregated; current new-campaign counts and selected scores are unavailable. The label-blind executable cohort freezes 1202 entries before scoring. The search distribution and locked-winner confirmation are reported separately in the appendix; a larger search may improve numerical matching without resolving historical identity.

3 What discrepancies can be explained

Numerics. For $c_e = 2\gamma w_e$, feasible q with $\|q_e\| \le c_e$ and $\|(B^T q)_i\| \le 1$ gives lower bound $D(q) = \langle B^T q, F\rangle$. We retain the best feasible initial dual before PDHG updates [Chambolle and Pock, 2011], and use relative gap at most 10^{-5} with capacity violations at most 10^{-8}. A diagonal formulation uses $K = \mathrm{diag}(\sqrt{w})B$ and maps its dual back to the original capacities. Independent small convex references validate the target. USPS $\gamma = 1, 1000$ audits use one frozen current F/W, historical-style IRLS, linear residual checks, continuation and bounded dual-flow candidates. Agreement between solvers alone is not certification; new arrays never impersonate missing historical artifacts.

Implementation and sources. Ding–Li LDA-Kmeans [Ding and Li, 2007], audited through a pinned ClustPy reference[1], can fail at singular S_w and can return a projection newer than its labels at the cap. We expose the projection actually used, count failures, compare 30/100/300 limits, and preserve the reference's behavior. A separately named PCA-preconditioned ridge LDA-Kmeans→SC adaptation uses $p_0 = \min(100, p, n - c, \mathrm{rank}(X_c))$, $r = \min(c - 1, p_0)$ and ridge $10^{-6}\mathrm{tr}(S_t)/p_0$, with matched controls. It does not recover historical LDA-SC. Paired one-shot SLPS [Hou et al., 2013] experiments separate $k = 5$ from the source's $k = 7$, alpha and dimension effects. Native 16-by16 EBSA MSRA retains all 1,799 rows and duplicates. Counts and good scores cannot establish historical pixel identity.

The repaired baseline implementations are available; new baseline result aggregation is pending. No additional benchmark result is inferred from implementation tests. Native 16-by16 MSRA and adapted LDA-to-SC remain separately named adaptations; exact historical LDA-SC remains unavailable.

4 Graph diagnostics and conditional extension

Incorrect W edges attract different classes, but that observation alone does not identify the main performance bottleneck. We measure A and W separately. Class-induced fragmentation now counts connected components of each $W[C, C]$, including isolates, excess fragments and largest-component fractions. Whole-graph coverage is retained under its accurate name. Deleting edges need not preserve $G = F$: on an eight-node cycle at $\gamma = 0.6$, deleting one edge destroys force cancellation. The sufficient incident-mass bound $2\gamma \max_i \sum_j w_{ij} \le 1$ is a separate statement.

The prior wrong-edge gate remains 0/4. New graph repair uses full preregistered ORL and cropped YaleB, independently of subset matching. G1 records complete SC/SSC error casebooks; G2 crosses anchors F_0, U with graphs W_0, W_a at matched total mass. G3 tests constrained oracle

[1] `https://github.com/collinleiber/ClustPy/blob/4ac9d00685efc7432800684f5bfac0f01af59172/clustpy/centroid/ldakmeans.py`; the observed port retains the BSD-3-Clause notice.

deletion/completion within a frozen, label-blind top-20-neighbor pool, against fixed oracle deletion plus blind random additions and separate randomized controls. Its $15\%/\gamma = 1000$ gate requires certified active shrinkage, at least 1-point paired ACC gain, nondecreasing NMI, 20% fewer excess fragments and positive conditional bootstrap intervals. Only that gate permits an EnSC graph adaptation [You et al., 2016]. No new Huber, capped-fusion or heat grids are launched.

G1–G3 are implemented and preregistered; no aggregated new graph-repair result is available. G4/G5 have not been promoted, and the four-benchmark target is unavailable. The earlier 0/4 wrong-edge gate is retained as a historical result; it is not replaced by this protocol.

The final practical target remains four-benchmark macro-average ACC gain of at least 2 points over the stronger fair SC/SSC capability baseline, without decreasing mean NMI relative to either. Both F_0-SC and U-SC belong to the fair-SC envelope. Capability-oracle and locked label-free operation are distinct; incomplete coverage leaves the four-benchmark conclusion unavailable. New rounding seeds on the same datasets assess conditional rounding stability, not unseen-data generalization.

5 Conclusion

SSC gives a clear convex refinement target, while clustering quality depends on representation, graph, solver and rounding choices. Our contribution is reproducible source/subset sensitivity and explicit numerical and implementation audits, including negative or unresolved evidence. Exact historical LDA-SC and several dataset identities remain unresolved. Code, complete records and commands are available at GitHub main.

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A Historical release and endpoint separation

The original scalar release and ACC-only selections remain historical records. Current joint selections are regenerated from complete seed groups; frozen measurements are not rewritten. Missing historical representations limit optimization re-audits on their exact arrays. The historical 72-cell identity/status matrix remains separate from newly measured candidates and adaptations. The original ACC-only closest-display and fixed-reference tables remain available in the committed historical release. The current joint 72-cell table below supersedes their display selection without rewriting any trial.

B Current authoritative results and coverage

B.1 Jointly reselected historical mean endpoint

Table 2 reports the complete 72-cell inventory. Values are percent. Every measured pair uses the same configuration and 50 batch-zero seeds; $\pm$ denotes sample SD. Missing cells retain their identity or method restriction. Historical scalar evidence is separate from new recoverable artifacts. The former ACC-only and fixed-reference tables remain in the committed historical release.

Table 2: Joint closest historical mean/sample-SD endpoint; unavailable cells are not filled with adapted measurements.

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
AR	K-means	36.30 ± 1.40	–	68.70 ± 3.00	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SC	41.60 ± 2.10	–	71.30 ± 2.60	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	PCA–K-means	39.80 ± 1.80	–	69.40 ± 2.80	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	PCA–SC	43.20 ± 1.70	–	70.30 ± 2.40	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LDA–K-means	40.40 ± 1.50	–	69.90 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LDA–SC	44.50 ± 1.30	–	70.80 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	LLC	48.70 ± 1.60	–	71.20 ± 2.40	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SLPS	49.20 ± 1.40	–	72.40 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
AR	SSC	51.30 ± 1.50	–	73.20 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
JAFFE	K-means	75.60 ± 1.80	71.97 ± 7.55	79.40 ± 0.80	81.97 ± 3.70	3.63
JAFFE; Y0, V; NOT_APPLICABLE; $D_{\text{mean}} = 3.10$.						
JAFFE	SC	76.10 ± 1.60	68.53 ± 11.15	80.20 ± 0.90	83.02 ± 5.72	7.57
JAFFE; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 5.20$.						
JAFFE	PCA–K-means	75.80 ± 1.50	73.20 ± 5.58	79.80 ± 0.80	82.95 ± 3.67	3.15
JAFFE; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 2.88$.						
JAFFE	PCA–SC	76.90 ± 1.80	72.30 ± 0.00	81.50 ± 1.30	82.86 ± 0.00	4.60
JAFFE; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 2.98$.						
JAFFE	LDA–K-means	76.50 ± 1.70	64.44 ± 9.19	82.10 ± 1.40	78.60 ± 5.08	12.06
JAFFE; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 7.78$.						
JAFFE	LDA–SC	77.40 ± 1.90	–	81.90 ± 0.90	–	–
No eligible complete measured configuration; historical restrictions remain						
JAFFE	LLC	78.60 ± 1.50	91.55 ± 0.00	82.50 ± 1.70	92.75 ± 0.00	12.95
JAFFE; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 11.60$.						
JAFFE	SLPS	79.50 ± 2.10	80.86 ± 5.77	83.10 ± 2.10	86.66 ± 3.34	3.56
JAFFE; Y0, one_shot, G; NOT_APPLICABLE; $D_{\text{mean}} = 2.46$.						
JAFFE	SSC	81.20 ± 1.60	74.75 ± 8.37	84.30 ± 1.60	85.60 ± 4.32	6.45
JAFFE; Y0, E13, $\gamma = 1e+06$, R; MAX_ITER_REACHED; $D_{\text{mean}} = 3.87$.						
ORL	K-means	60.50 ± 1.80	61.46 ± 2.85	80.30 ± 1.80	82.76 ± 1.17	2.46
ORL_att92; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 1.71$.						
ORL	SC	72.70 ± 2.30	72.73 ± 3.15	85.80 ± 1.90	87.49 ± 1.22	1.69
ORL_att92; Y0, V; NOT_APPLICABLE; $D_{\text{mean}} = 0.86$.						
ORL	PCA–K-means	64.50 ± 2.30	63.08 ± 2.26	80.60 ± 1.60	82.34 ± 0.90	1.74
ORL_att92; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 1.58$.						

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
ORL	PCA–SC	67.80 ± 2.10	67.57 ± 2.92	86.30 ± 1.40	85.81 ± 1.27	0.49
ORL_att92; Y0, V; NOT_APPLICABLE; $D_{\mathrm{mean}} = 0.36$.						
ORL	LDA–K-means	65.60 ± 2.60	56.31 ± 4.14	81.10 ± 2.10	81.00 ± 1.65	9.29
ORL_att92; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\mathrm{mean}} = 4.70$.						
ORL	LDA–SC	68.30 ± 2.40	–	86.80 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
ORL	LLC	71.50 ± 2.20	81.07 ± 0.61	84.90 ± 1.50	89.92 ± 0.27	9.57
ORL_att92; Y0, YU_SHI; NOT_APPLICABLE; $D_{\mathrm{mean}} = 7.29$.						
ORL	SLPS	74.20 ± 1.80	73.66 ± 2.13	87.20 ± 1.80	87.71 ± 0.91	0.54
ORL_att92; Y0, one_shot, G; NOT_APPLICABLE; $D_{\mathrm{mean}} = 0.53$.						
ORL	SSC	76.00 ± 1.60	74.63 ± 0.64	88.60 ± 1.50	89.71 ± 0.23	1.37
ORL_att92; Y0, E8, $\gamma = 1000$, V; ITERATE_CONVERGED_UNCERTIFIED; $D_{\mathrm{mean}} = 1.24$.						
UMIST	K-means	41.30 ± 1.60	–	64.40 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SC	52.20 ± 1.40	–	72.10 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	PCA–K-means	48.80 ± 1.70	–	68.20 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	PCA–SC	54.10 ± 1.90	–	72.90 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LDA–K-means	49.70 ± 1.80	–	68.80 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LDA–SC	54.70 ± 1.50	–	74.10 ± 2.00	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	LLC	63.30 ± 1.80	–	77.30 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SLPS	70.40 ± 1.60	–	82.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
UMIST	SSC	71.10 ± 1.80	–	84.10 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	K-means	41.70 ± 1.10	–	58.60 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SC	43.60 ± 1.50	–	59.70 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	PCA–K-means	42.40 ± 1.50	–	59.10 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	PCA–SC	44.30 ± 1.80	–	60.60 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LDA–K-means	42.90 ± 1.70	–	59.80 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LDA–SC	45.10 ± 1.40	–	61.30 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	LLC	40.70 ± 1.80	–	61.40 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SLPS	48.50 ± 1.90	–	63.60 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
BinAlpha	SSC	49.40 ± 1.30	–	64.10 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	K-means	46.20 ± 1.70	–	56.70 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SC	52.30 ± 1.80	–	70.00 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	PCA–K-means	56.00 ± 1.90	–	60.30 ± 1.50	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	PCA–SC	54.90 ± 1.60	–	72.40 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LDA–K-means	56.40 ± 1.80	–	61.10 ± 1.90	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LDA–SC	55.10 ± 1.70	–	73.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	LLC	48.10 ± 1.40	–	66.20 ± 1.60	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SLPS	60.30 ± 1.30	–	69.60 ± 1.80	–	–
No eligible complete measured configuration; historical restrictions remain						
MSRA50	SSC	63.20 ± 1.10	–	72.20 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						

Data	Method	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
YaleB	K-means	14.40 ± 1.50	9.74 ± 0.46	17.30 ± 1.50	13.25 ± 0.59	4.66
YaleB_dengcai32; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 4.35$.						
YaleB	SC	34.80 ± 1.40	37.39 ± 2.22	55.60 ± 1.60	52.96 ± 1.00	2.64
YaleB_dengcai32; Y3, R; NOT_APPLICABLE; $D_{\text{mean}} = 2.61$.						
YaleB	PCA–K-means	24.90 ± 1.80	9.85 ± 0.55	26.70 ± 1.80	13.32 ± 0.66	15.05
YaleB_dengcai32; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 14.22$.						
YaleB	PCA–SC	36.80 ± 2.00	30.92 ± 0.54	38.60 ± 1.50	38.86 ± 0.95	5.88
YaleB_dengcai32; Y1, G; NOT_APPLICABLE; $D_{\text{mean}} = 3.07$.						
YaleB	LDA–K-means	26.10 ± 1.80	9.80 ± 0.85	29.40 ± 1.60	13.79 ± 1.15	16.30
YaleB_dengcai32; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 15.96$.						
YaleB	LDA–SC	38.20 ± 1.60	–	39.90 ± 1.40	–	–
No eligible complete measured configuration; historical restrictions remain						
YaleB	LLC	38.20 ± 1.50	32.35 ± 0.05	34.10 ± 1.30	39.68 ± 0.11	5.85
YaleB_dengcai32; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 5.72$.						
YaleB	SLPS	47.30 ± 1.70	13.52 ± 0.78	41.40 ± 1.60	19.44 ± 0.69	33.78
YaleB_dengcai32; Y0, one_shot, G; NOT_APPLICABLE; $D_{\text{mean}} = 27.87$.						
YaleB	SSC	49.80 ± 1.60	42.64 ± 0.61	46.80 ± 1.30	52.19 ± 0.31	7.16
YaleB_dengcai32; Y3, E8, $\gamma = 1e+06$, G; MAX_ITER_REACHED; $D_{\text{mean}} = 6.27$.						
USPS	K-means	65.40 ± 1.70	66.06 ± 2.17	67.30 ± 1.80	61.04 ± 0.87	6.26
USPS; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 3.46$.						
USPS	SC	64.30 ± 1.40	61.74 ± 8.09	69.50 ± 1.60	74.94 ± 4.49	5.44
USPS; Y0, R; NOT_APPLICABLE; $D_{\text{mean}} = 4.00$.						
USPS	PCA–K-means	69.40 ± 1.80	65.25 ± 2.85	73.10 ± 1.90	61.10 ± 0.95	12.00
USPS; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 8.08$.						
USPS	PCA–SC	69.10 ± 1.50	65.53 ± 3.26	74.20 ± 1.80	78.92 ± 2.12	4.72
USPS; Y0, G; NOT_APPLICABLE; $D_{\text{mean}} = 4.15$.						
USPS	LDA–K-means	70.10 ± 1.30	58.61 ± 6.33	75.10 ± 1.60	60.29 ± 4.21	14.81
USPS; Y0, FULL_REFIT; NOT_APPLICABLE; $D_{\text{mean}} = 13.15$.						
USPS	LDA–SC	70.40 ± 1.50	–	75.40 ± 1.70	–	–
No eligible complete measured configuration; historical restrictions remain						
USPS	LLC	63.90 ± 1.70	65.27 ± 0.01	67.50 ± 1.50	72.76 ± 0.01	5.26
USPS; Y0, YU_SHI; NOT_APPLICABLE; $D_{\text{mean}} = 3.32$.						
USPS	SLPS	71.40 ± 1.60	70.81 ± 8.33	76.50 ± 1.80	76.30 ± 2.47	0.59
USPS; Y0, one_shot, R; NOT_APPLICABLE; $D_{\text{mean}} = 0.39$.						
USPS	SSC	75.50 ± 1.90	73.87 ± 7.53	79.80 ± 1.60	82.84 ± 3.96	3.04
USPS; Y0, E8, $\gamma = 1000$, R; ITERATE_CONVERGED_UNCERTIFIED; $D_{\text{mean}} = 2.34$.						

B.2 Native-objective-selected endpoint

Within one fixed G, native K-means SSE selects one of50 assignments. This is a single assignment with no across-run SD. It does not compete with the mean endpoint. Full method-specific endpoint records are in `results/submission_refresh/endpoint_comparison.csv`; LDA cross-subspace native selection is unavailable.

SSC data	ACC	NMI	$D_{\max}$	D_{mean}	Seed	Fit
AR	–	–	–	–	–	–
JAFFE	95.31	94.09	14.11	11.95	13	T
ORL	75.00	88.60	1.00	0.50	40	I
UMIST	–	–	–	–	–	–
BinAlpha	–	–	–	–	–	–
MSRA50	–	–	–	–	–	–
YaleB	43.33	54.15	7.35	6.91	45	I
USPS	79.62	85.38	5.58	4.85	38	I

B.3 Dataset-subset matching: both endpoints and eligibility pools

The following rows use complete evaluated configurations only. Paper values are fixed SSC targets. R/V/G denote random, vanilla K-means++ and sklearn greedy K-means++; each is one Lloyd start. Specification-compatible dimensions/counts do not establish historical pixel identity. Adapted rows retain explicit source/resolution/preparation differences. Diagnostic parents cannot win either pool.

Mean/sample-SD over 50 fixed-G roundings

Data / pool	Selected candidate	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
AR / no complete eligible result	Unavailable	51.30 ± 1.50	–	73.20 ± 1.40	–	–
JAFFE / no complete eligible result	Unavailable	81.20 ± 1.60	–	84.30 ± 1.60	–	–
ORL / no complete eligible result	Unavailable	76.00 ± 1.60	–	88.60 ± 1.50	–	–
UMIST / no complete eligible result	Unavailable	71.10 ± 1.80	–	84.10 ± 1.50	–	–
BinAlpha / no complete eligible result	Unavailable	49.40 ± 1.30	–	64.10 ± 1.40	–	–
MSRA50 / no complete eligible result	Unavailable	63.20 ± 1.10	–	72.20 ± 1.40	–	–
YaleB / no complete eligible result	Unavailable	49.80 ± 1.60	–	46.80 ± 1.30	–	–
USPS / no complete eligible result	Unavailable	75.50 ± 1.90	–	79.80 ± 1.60	–	–

Single native-best assignment among 50 fixed-G roundings

Data / pool	Selected candidate	Paper ACC	Our ACC	Paper NMI	Our NMI	$D_{\max}$
AR / no complete eligible result	Unavailable	51.30 ± 1.50	–	73.20 ± 1.40	–	–
JAFFE / no complete eligible result	Unavailable	81.20 ± 1.60	–	84.30 ± 1.60	–	–
ORL / no complete eligible result	Unavailable	76.00 ± 1.60	–	88.60 ± 1.50	–	–
UMIST / no complete eligible result	Unavailable	71.10 ± 1.80	–	84.10 ± 1.50	–	–
BinAlpha / no complete eligible result	Unavailable	49.40 ± 1.30	–	64.10 ± 1.40	–	–
MSRA50 / no complete eligible result	Unavailable	63.20 ± 1.10	–	72.20 ± 1.40	–	–
YaleB / no complete eligible result	Unavailable	49.80 ± 1.60	–	46.80 ± 1.30	–	–
USPS / no complete eligible result	Unavailable	75.50 ± 1.90	–	79.80 ± 1.60	–	–

B.4 Additional rounding on locked winners

No completed additional-rounding record is currently aggregated. Winners, if available, remain locked; no stability result is inferred.

B.5 Complete baseline repair endpoints

Baseline result aggregation is pending or contains no complete endpoint. Implementation tests do not establish benchmark performance.

B.6 Coverage, solver qualification and cost

Data	Intended aliases	Unique inputs	Measured inputs	Complete configurations
AR	3537	3533	NA	NA
JAFFE	1	1	NA	NA
ORL	1	1	NA	NA
UMIST	1073	971	NA	NA
BinAlpha	104	104	NA	NA
MSRA50	1	1	NA	NA
YaleB	16	16	NA	NA
USPS	3	3	NA	NA

Shared ledger charged 0.1679 CPU-core hours as of this aggregation. Limit 36, with 2 reserved for final checks; at most 2 single-thread workers and 8 GiB aggregate RSS target. Active reservations: 1. This is a snapshot, not a prediction of final resource use.

Generated-input provenance. Tables and figures are generated from the authoritative committed CSV/JSON records only. The input hash manifest is `results/submission_refresh/report_input_manifest.json`. Generated snapshot date: 2026-11-10.

C Earlier numerical and mechanism findings

The following records describe the earlier campaign. Its measured CPU costs and 0/4 gate are historical; the current combined campaign uses the separate shared ledger above. The truncated fusion penalty already appears in PRclust [Pan et al., 2013, Wu et al., 2016]; robust pairwise fusion is also central to Shah–Koltun RCC [Shah and Koltun, 2017]. These works differ from Sun et al.'s resistant convex clustering. We claim no invention of those penalties.

C.1 Certificate definitions

For radial Huber, $D_\delta(q) = \langle B^T q, F\rangle - (\delta/2)\sum_i \|(B^T q)_i\|^2/a_i$. Edge and node balls apply; squared fidelity removes node balls. An inward capacity-ratio scaling recovers feasible q. Best primal means an actual stored G; objective, array and clustering metrics are never mixed across iterates. Negative primal-dual gaps beyond tolerance are errors. The simple $G = F$ sufficient region uses $b = B^T(2wu)$, with unit directions on noncoincident F-edges and zero otherwise, and $\gamma_0 = 1/\max_i \|b_i\|$. Coincident edges need not yield the maximal onset. Infinity is asserted only for zero-TV analytic cases, not a finite search reaching a large gamma. All 88 historical verified claims were audited: 'ANALYTIC_G_EQUALS_F': 45, 'RECORDED_OBJECTIVE_GAP_CERTIFIED': 37, 'LEGACY_RETURNED_ITERATE_UNCERTIFIED': 6. Direct analytic G=F checks and certification using a recorded scalar primal are distinguished; the latter is not recovery of an unavailable historical G array. Every old status remains in the audit CSV. The numerical amendment retains cold runs and adds an exact-objective-accepted IRLS primal initializer (at most 300 iterations for n<=3000, otherwise 30; at most 60 seconds). Failure falls back to the cold start. It changes neither target, graph, parameter grid nor certificate threshold. The code/config amendment is explicit; development data were already inspected, so later results are not independent held-out validation. Capped fusion uses active-edge block minimization and two initializations. Inner convex gap certification is not a global certificate for the nonconvex objective. Fixed-point, truncation, monotonicity rejection and missing second-start states remain explicit. Tiny-graph state enumeration supplies an independent global reference only for those tiny problems.

C.2 Candidate mechanisms

Radial Huber is convex and orthogonally invariant. Along $G(t) = (1-t)F + t1m^T$, graph TV decreases linearly while near-zero fidelity increases quadratically, so nontrivial F is not optimal for positive gamma/delta. This proves activation, not better ACC. The bound $0 \le r - \phi_\delta(r) \le \delta/2$ gives original-objective error at most $n\delta/2$ plus a certified absolute solver gap. Bounded radial fidelity influence is not a transferred breakdown theorem [Sun et al., 2025]. Heat features use $H_t = e^{-tL/2}$ with $H_t H_t^T = K_t = e^{-tL}$; rows of K_t correspond instead to time 2t distances. Duhamel bounds kernel operator change by $t\|L - L'\|_2$ and each squared-distance change by $2t\|L - L'\|_2$. A kNN boundary gap above $4t\|L - L'\|_2$ cannot cross under that perturbation. Full-spectrum random projection is approximate and does not automatically inherit this exact bound. Neither stability nor a large diffusion time guarantees labels [Coifman and Lafon, 2006, Gasteiger et al., 2019].

C.3 Evaluation and resource limits

The old robustness table did not rerun SSC, and used the wrong USPS pixel domain. It is retained with an invalidity annotation. Corrected perturbations declare [-1,1] for USPS, preserve zero-change identity and rerun the full corresponding pipeline. ARI/Jaccard alone do not measure class accuracy. All primary K-means endpoints use 50 seeds, one greedy++ initialization and sample SD. Paired bootstrap uses seed 20261108 and 2000 resamples. Perturbation seed, not each correlated rounding, is an independent unit. Missing four-benchmark coverage leaves the formal average NA; the two YaleB sources are not independent benchmarks. New computation is capped at 36 CPU-core hours, two workers and a target 8 GiB total RSS, with no GPU or paid resources. PDHG has at most 30000 iterations per fit and 10/30-minute small/large timeouts. Runtime, memory and all retained solver versions are in the resource ledger. Failure never becomes zero ACC. Native prclust 1.3 was compiled at a pinned source revision and executed on a five-point sanity check; this is not a benchmark result. New optimization modules are from-scratch formula implementations. Scientific snapshots, data restrictions, exact source revisions and clone verification are indexed in docs/final. The squared control uses $\sum_i \|g_i - f_i\|^2/(2s_F)$. Multiplying by $2s_F$ maps its linear-fusion coefficient to ordered-pair Equation 7 as $\gamma_{E7} = 2s_F\gamma$. Equal numerical gamma values across fidelities do not imply equal regularization strength; separate searches are required. For N1 quadratic scale delta, replace s_F by delta. Measured fit/audit CPU is 2.623 core-hours; the conservative two-worker session envelope is 3.822, below 36. Largest recorded per-fit peak RSS is 597.7 MiB. Aggregate RSS was spot-checked, not continuously enforced.

C.4 Separate A and W audit

Label-assisted fractions and AUROC on the newly generated Y0 representations. W length is measured in F, with cross-class edges positive; values below .5 mean longer edges do not identify errors. A native-input length AUROC is not asserted.

Dataset	Graph	cross edges (%)	cross mass (%)	length AUROC
ORL_att92	A	22.31	11.99	–
ORL_att92	W	17.01	4.65	0.830
JAFFE	A	3.55	0.21	–
JAFFE	W	6.87	7.13	0.556
YaleB_cropped2414	A	58.88	53.60	–
YaleB_cropped2414	W	63.24	70.23	0.380
USPS	A	5.04	2.32	–
USPS	W	5.17	2.00	0.758
YaleB_dengcai32	A	51.95	46.64	–
YaleB_dengcai32	W	54.28	63.43	0.360

C.5 Synthetic mechanism checks

Synthetic seeds 100–109 are independent instances; each has 50 conditional K-means seeds. Parameters and distance scales are frozen from the unpolluted instance. The full 2-by-2 and static-screening controls are retained, including uncertified inner solves. This intentionally easy geometry can show damage avoidance but is weak evidence of added clustering capacity.

Scenario	Method	ACC	NMI	qualified / 10
anchor_0.2	SC	100.000	100.000	10
anchor_0.2	SSC	100.000	100.000	10
anchor_0.2	radial_capped	100.000	100.000	10
anchor_0.2	squared_linear	100.000	100.000	10
anchor_0.2	static_screen	100.000	100.000	10
bridge_0.1	SC	100.000	100.000	10
bridge_0.1	SSC	99.889	99.555	10
bridge_0.1	radial_capped	100.000	100.000	10
bridge_0.1	squared_linear	100.000	100.000	10
bridge_0.1	static_screen	100.000	100.000	10
imbalanced_bridge	SC	100.000	100.000	10
imbalanced_bridge	SSC	99.156	96.725	10
imbalanced_bridge	radial_capped	100.000	100.000	10
imbalanced_bridge	squared_linear	100.000	100.000	10
imbalanced_bridge	static_screen	100.000	100.000	10

Bridge contamination slightly lowers SSC ACC/NMI; capping, static deletion and squared fidelity all avoid this damage, while SC is already perfect. Capping is therefore neither necessary nor shown to improve beyond SC. At 20% anchor contamination all methods have 100% ACC/NMI, yet capping can worsen representation MSE by removing useful correction. The radial-Huber primary pilot changes MSE slightly without label improvement; its full real-data search is not promoted.

C.6 Tiny global reference and initialization dependence

For the documented five-point graph, 1024 active states are enumerated, 960 excluded by a nonnegative constant lower bound, and 64 convex states independently certified. Original primal/dual are approximately 8; capped global primal/dual are 3. The local capped algorithm returns objective 3 (ACC/NMI 100%) from F, but objective 8 (ACC 60%, NMI 0) from original SSC. Global certification comes solely from enumeration on this tiny graph.

C.7 Secondary heat ablation

The 80-row geometry pilot shows higher projected-heat kNN stability, not label improvement. Primary t=4, r=128 uses the same A, full-spectrum action, fixed projection seed, hard-dimension and W-only controls. Below, each cell is ACC/NMI (percent); common gamma=1 and own $2\gamma_0$ are locked without labels. C/U denote certified/uncertified in that order. Four original gamma=1 controls are reused from Table 2; 28 new fits and their actual gamma/costs are in the scalar release. N2 changes F and cannot substitute for the same-original-F main goal.

Data	Feature	SC ACC/NMI	gamma=1 ACC/NMI	own $2\gamma_0$ ACC/NMI	status
ORL	original	78.33/89.87	78.39/89.74	78.40/89.59	C/C
ORL	hard128	47.97/67.93	44.39/71.72	47.82/67.91	C/C
ORL	heat128	75.19/87.96	77.38/89.07	76.45/88.68	C/C
ORL	W-only	78.33/89.87	77.69/90.05	77.97/89.66	C/C
JAFFE	original	95.31/94.13	95.10/94.06	95.31/94.13	C/C
JAFFE	hard128	23.65/29.29	10.80/0.00	22.59/27.09	C/C
JAFFE	heat128	51.55/60.41	55.42/64.90	53.62/62.34	C/C
JAFFE	W-only	95.31/94.13	95.21/94.23	95.31/94.14	C/C
YaleB	original	21.78/27.73	21.55/27.49	21.79/27.70	C/C
YaleB	hard128	15.81/24.15	14.64/24.89	15.97/24.51	C/C
YaleB	heat128	19.75/24.07	17.05/29.46	19.42/24.12	C/C
YaleB	W-only	21.78/27.73	21.87/27.64	21.83/27.69	C/C
USPS	original	65.60/80.31	65.40/80.11	65.55/80.42	U/U
USPS	hard128	30.32/33.81	26.84/27.72	29.95/33.32	U/C
USPS	heat128	17.23/3.44	17.38/4.47	17.39/3.55	U/C
USPS	W-only	65.60/80.31	63.71/76.77	65.41/80.55	U/C

The exact kernel bound applies before projection. Fixed-subset exact-kernel errors and restricted-candidate kNN recall are separate diagnostics, not full-graph guarantees. No t/r/seed tuning or projection sensitivity is claimed.

C.8 Corrected full-pipeline image conditions

Four fixed image conditions reconstruct A/F/W/SSC at clean gamma=1; USPS uses [-1,1] with full-span brightness steps. Ranges below are descriptive changes in percentage points against each clean method. Conditions are not independent random replicates; no inferential or adaptive-selection claim is made. USPS SSC remains uncertified.

Data	SC ΔACC	SSC ΔACC	SC ΔNMI	SSC ΔNMI
ORL	[-0.06,1.35]	[0.00,1.76]	[-0.23,0.37]	[0.00,0.84]
JAFFE	[0.00,0.00]	[0.00,0.21]	[0.00,0.00]	[0.00,0.08]
YaleB	[0.08,0.18]	[0.03,0.61]	[0.08,0.20]	[0.03,0.42]
USPS	[-0.36,0.09]	[-0.05,0.36]	[-0.32,0.48]	[0.05,0.75]